

Propositional Logic Series

Episode 2: Worksheet

Instructions

This worksheet is about arguments, premises, conclusions, standard form, deductive and inductive arguments, and the first idea of validity. Most examples are mathematical.

Part A. Truth-Apt Expressions

For each item, write whether it is truth-apt in the strict logical sense.

1. 24 is divisible by 4.
2. Is 24 divisible by 4?
3. Every integer divisible by 10 is divisible by 5.
4. Prove that $\sqrt{2}$ is irrational.
5. 7 is a prime number.
6. Choose a natural number n .
7. The set $\{1, 2\}$ is a subset of $\{1, 2, 3\}$.
8. Let us solve the equation.

Part B. Arguments and Nonarguments

For each passage, decide whether it is an argument or not an argument. If it is an argument, briefly identify what is being supported. If it is not an argument, briefly say why.

1. If an integer is divisible by 6, then it is divisible by 3. The integer 120 is divisible by 6. Therefore, 120 is divisible by 3.
2. 2 is prime. 3 is prime. 5 is prime. 7 is prime.
3. Since $A \subseteq B$ and $x \in A$, it follows that $x \in B$.
4. Let n be an even integer. Prove that n^2 is even.
5. The number 48 is even because it is divisible by 2. In this context, assume that we already know that 48 is even and are asking why.

6. 10 is divisible by 5 and by 2. 20 is divisible by 5 and by 2. 30 is divisible by 5 and by 2. Therefore, every natural number divisible by 5 is divisible by 2.
7. The function f is differentiable at a . Every differentiable function is continuous at the point where it is differentiable. Hence f is continuous at a .
8. The empty set has no elements. The set $\{1, 2, 3\}$ has three elements. The set of natural numbers is infinite.
9. If the server is down, then the website is unavailable. The server is down. So the website is unavailable.
10. The ground is wet because it rained last night. In this context, assume that we are already looking at the wet ground and asking why it is wet.

Part C. Premises and Conclusions

For each argument, identify the premises and the conclusion.

1. If an integer is divisible by 6, then it is divisible by 3. The integer 120 is divisible by 6. Therefore, 120 is divisible by 3.
2. Since every differentiable function is continuous, and f is differentiable at a , f is continuous at a .
3. 15 is divisible by 5. If a number is divisible by 5, then it has last digit 0 or 5. So 15 has last digit 0 or 5.
4. $A \subseteq B$, and $x \in A$. Hence $x \in B$.
5. The server is down. If the server is down, the website is unavailable. Therefore, the website is unavailable.
6. The triangle has three equal sides. Every triangle with three equal sides is equilateral. Thus, the triangle is equilateral.
7. n is even because n is divisible by 4.
8. Since p is prime and $p > 2$, p is odd.

Part D. Standard Form

Put each passage into standard form. Write the premises first and the conclusion last.

1. 72 is divisible by 8, because $72 = 8 \cdot 9$.
2. $x \in C$, since $A \subseteq C$ and $x \in A$.
3. The function f is continuous at a . After all, f is differentiable at a , and every differentiable function is continuous.

4. Therefore, 36 is divisible by 3. The reason is that 36 is divisible by 6, and every number divisible by 6 is divisible by 3.
5. The program crashes, because the file is corrupted and corrupted files cause the program to crash.
6. m is not odd. For m is even, and no even integer is odd.

Part E. Argument, Explanation, or Mere List

Classify each passage as an argument, an explanation, or a mere list. Give a short reason.

1. 2 is prime. 3 is prime. 5 is prime. 7 is prime.
2. The number 48 is even because it is divisible by 2. In this context, assume that we already know that 48 is even and are asking why.
3. 48 is divisible by 2. Therefore, 48 is even.
4. The ground is wet because it rained last night. In this context, assume that we are already looking at the wet ground and asking why it is wet.
5. If $A \subseteq B$, then every element of A is an element of B . Also, $x \in A$. Therefore, $x \in B$.
6. 10 is a natural number. π is irrational. The empty set has no elements.

Part F. Deductive or Inductive

Classify each argument as deductive or inductive. Give a short reason.

1. If an integer is divisible by 12, then it is divisible by 3. The integer 84 is divisible by 12. Therefore, 84 is divisible by 3.
2. 10 is divisible by 5 and by 2. 20 is divisible by 5 and by 2. 30 is divisible by 5 and by 2. Therefore, every natural number divisible by 5 is divisible by 2.
3. 2 is prime, 3 is prime, 5 is prime, and 7 is prime. Therefore, all odd natural numbers greater than 1 are prime.
4. If $A = B$, then $A \subseteq B$. We have $A = B$. Therefore, $A \subseteq B$.
5. The first ten values of this sequence are positive. Therefore, every term of the sequence is positive.
6. If x is a multiple of 4, then x is even. x is a multiple of 4. Therefore, x is even.
7. The last several examples of this polynomial had integer roots. Therefore, this polynomial probably has an integer root.
8. If f is differentiable at a , then f is continuous at a . f is differentiable at a . Therefore, f is continuous at a .

Part G. Validity Preview

For each deductive argument, say whether it is valid or invalid. If it is invalid, give a counterexample or describe a possible case where all premises are true and the conclusion is false.

1. If n is divisible by 4, then n is even. 20 is divisible by 4. Therefore, 20 is even.
2. Every integer divisible by 4 is even. 6 is even. Therefore, 6 is divisible by 4.
3. If f is differentiable at a , then f is continuous at a . The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x|$ is continuous at 0. Therefore, f is differentiable at 0. (Here, $\mathbb{R}$ denotes the set of all real numbers.)
4. $A \subseteq B$ and $x \in A$. Therefore, $x \in B$.

Sources and Note

The conceptual background for this worksheet is informed by P. D. Magnus, Tim Button, Robert Trueman, and Richard Zach, *forall x: Calgary: An Introduction to Formal Logic*, with contributions by J. Robert Loftis and Aaron Thomas-Bolduc.

Official page: <https://forallx.openlogicproject.org/>

The examples and exercises in this worksheet are original teaching materials prepared for the Propositional Logic Series. They are not copied from the textbook.